

Alexander Turco

Doctoral Student, PhD in Medical Biophysics, University of Toronto

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Education

University of Toronto: PhD in Medical Biophysics

Sep 2024 - Present

Thesis Supervisor: Dr. Sushant Kumar

- PhD Thesis Title: Repetitive Genomic Elements as Multi-Scale Vulnerabilities in Cancer: Examining DNA Structure, Sequence, and Protein-Level Repetition

McMaster University: Honours BSc in Biology, Research Specialization

Sep 2019 - Dec 2023

Thesis Supervisors: Dr. G. Brian Golding, Dr. Rosa da Silva

- Honours Thesis Title (Dr. Golding): Estimating Evolutionary Parameters for Protein Low Complexity Regions using an Approximate Bayesian Computation
- Honours Thesis Title (Dr. da Silva): Cells at War: The Playfulness of Game-Based Learning

Publications & Preprints

Turco, A., Boev, N. & Kumar, S.

2026

Diversity and genomic organization of non-B DNA motifs in haplotype-resolved human genome assemblies. [bioRxiv](#)

- [doi:10.64898/2026.03.05.709836](https://doi.org/10.64898/2026.03.05.709836)

Specialized Skills

Programming Languages and High Performance Computing: R, Python, C++, Git/GitHub, Unix/Linux, Bio-Conductor, bash, ComputeCanada, L^AT_EX

Genomic Tools & Methods: SPAdes, Kraken2, BLAST, FastP, sam/bamtools, GSEA, DESeq2, PCA, BWA, tidyverse, ggplot, dplyr, Dorado (Nanopore), Remora (Nanopore), minimap2, Verkko, hifiasm, IGV, PyTorch, Scanpy, AnnData

Soft Skills: Collaboration, communication, detail-oriented, highly organized, creative problem solver, highly adaptive

Research Experience

Research Technician I - Kumar Lab, Computational Cancer Genomics

Jan 2024 - August 2024

Princess Margaret Cancer Research Centre, University Health Network

Toronto, ON, Canada

- Explored alternative (non-B) DNA structures using long-read sequencing data.
- Built a workflow to process (basecall + align) raw long read sequencing data from the Human Genomic Structural Variation Consortium (HGSVC) and extract translocation time metrics (time it takes for a DNA base to pass through sequencing nanopore).
- Developed a machine learning model to detect the presence of alternative (non-B) DNA structures based on extracted translocation times.
- Examined the landscape of potential non-B DNA structures across the genome using high quality, phased, telomere-to-telomere genome assemblies.

Research Student - Kumar Lab, Computational Cancer Genomics May 2023 - August 2023
Princess Margaret Cancer Research Centre, University Health Network *Toronto, ON, Canada*

- Explored sex differences in gene expression across twelve human cancers to elucidate genetic interactions that selectively kill cancerous cells (synthetic lethal interaction).
- Developed a bioinformatic pipeline to analyze gene expression (RNA-seq) data from The Cancer Genome Atlas (TCGA), specifically focusing on determining differentially expressed genes that interact in a synthetic lethal manner.
- Created detailed documentation on operating procedures for computational pipeline.
- Concisely communicated scientific research to field specific and public audiences.

Research Student - Golding Lab, Bioinformatics and Molecular Evolution May 2022 - April 2023
Department of Biology, McMaster University *Hamilton, ON, Canada*

- Explored the microbial composition of freshwater algal bloom sites across Ontario (summer project), as well as the evolution of protein low complexity regions (undergraduate thesis).
- Utilized bioinformatic tools and experimental design related to data visualization, genomic data analysis, phylogenetics, and molecular evolution.
- Analyzed and manipulated 16s rRNA amplicon sequence data collected by the Ministry of Environment and Climate Change (MOECC) to understand the toxicity of algal blooms.
- Developed a C++ program to simulate the evolution of protein low complexity regions as part of a step in an Approximate Bayesian Computation, in order to predict parameters that accurately describe the evolution of these regions.
- Comprehensive training in bioinformatic software and high performance computing such as R, Python, and Unix.
- Created detailed documentation describing background information, methods, and results.
- Concisely communicated scientific research through oral and poster presentations at two conferences.

Research Student - da Silva Lab, Pedagogy and Science Education May 2022 - April 2023
Department of Biology, McMaster University *Hamilton, ON, Canada*

- Explored the impacts of bringing game-based learning into university classrooms, through the development of a biological video game called "Cells at War".
- Collaborated with artists, designers, programmers, musicians, and scientists across the globe to conceptualize, design, and build an educational tool to teach first year students core cellular and molecular biology concepts.
- Provided biological expertise, and applied critical thinking strategies to synchronize scientific facts with the creative game design process.
- Created student feedback survey and analyzed results to better understand how video games improve student engagement and motivation.
- Communicated scientific research through oral presentations at two conferences as well as a full research paper highlighting student perceptions on game-based learning.

Presentations and Conferences

Invited Speaker Sep 2025

Seminar speaker @ Buffalo Evolutionary and Anthropological Genomics Lab (BEAGL) *Buffalo, NY, USA*

- Invited to speak about non-B DNA structures and their role in evolution and genomic instability at the University at Buffalo

Oral Presentation Aug 2025

Human Genomic Structural Variation Consortium (HGSVC) *Virtual*

- Presented work on non-B DNA structures in high quality haplotype-resolved human assemblies. The HGSVC includes a diverse group of scientists around the world actively working to assemble, annotate, and explore high quality, diverse genome assemblies.

Poster Presentation

May 2025

Medical Biophysics Graduate Student Symposium, University of Toronto

Toronto, ON, Canada

- The Medical Biophysics Graduate Symposium brings together graduate students and faculty in the department of Medical Biophysics for a day to showcase research.

Poster Presentation

April 2025

Toronto DNA Replication and Repair Symposium, University of Toronto

Toronto, ON, Canada

- The [Toronto DNA Replication and Repair Symposium](#) aims to connect researchers in the field of genome integrity, DNA replication and repair across departments and institutes.

Conference Attendee

September 2024

Telomere-to-Telomere Face-to-Face Conference, University of California Santa Cruz

Santa Cruz, CA, USA

- The [Telomere-to-Telomere \(T2T\) consortium](#) is an open, community-based effort to generate the first complete assembly of a human genome.

Oral Presentation

August 2023

University Health Network Summer Training and Research Program

Toronto, ON, Canada

- 3 minute thesis virtual presentation.

Oral Presentation

July 2023

The Western Conference on Science Education

London, ON, Canada

- A STEAM game-based learning framework: Maximizing integrated and immersive learning in the classroom.
- Presented by supervising professor Dr. Rosa da Silva
- Conference publication available in [The Western Conference on Science Education Journal, 2023](#)

Oral Presentation

April 2023

Biology Undergraduate Symposium, McMaster University

Hamilton, ON, Canada

- Undergraduate thesis presentation in computational biology.

Oral Presentation

April 2023

Biology Undergraduate Symposium, McMaster University

Hamilton, ON, Canada

- Undergraduate thesis presentation in science education.

Poster Presentation

October 2022

MacWater Challenges in Water Monitoring Conference

Hamilton, ON, Canada

- Poster presentation on 16s rRNA sequencing analysis of harmful algal blooms in Ontario.

Awards & Honors

MBP Excellence OSOTF PMH Award

Ontario Cancer Institute & Princess Margaret Cancer Centre \$5,000.00

Aug 2025

Canada Graduate Scholarship Masters (CGS-M)

Canadian Institutes of Health Research \$27,000.00

May 2025

Best Poster Award

Toronto DNA Replication and Repair Symposium, University of Toronto, \$100.00

April 2025

Oral Presentation Award in Computational Biology

Biology Undergraduate Symposium, McMaster University

April 2023

Oral Presentation Award in Science Education

Biology Undergraduate Symposium, McMaster University

April 2023

3rd Place Abstract Award

MacWater Challenges in Water Monitoring Conference, McMaster University

October 2022

Research stipend for the creation of Cells at War: A Biological Video Game

Co-operative Education and Work-Integrated Learning Canada (CEWIL), \$6000.00 *Sept 2021 - Dec 2023*